

Video Codecs Towards 2030

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01

Video Coding Standard Beyond
H.266/VVC

Enhanced Compression Model (ECM)

- In April 2021, shortly after VVC version 1 was finalized, JVET-U0100 showed 11.5% coding performance gain vs. VVC based on “known” coding tools
- In July 2021, enhanced compression model (ECM) software platform was established to facilitate exploration of coding tools
- From the first version (ECM-1.0, July 2021) to the last version (ECM-20.0, July 2026), coding performance steadily improves
- ECM-20.0 achieves 27.6% gain vs. VVC, with 10x encoder and 14x decoder runtime
- Based on hybrid block-based video coding framework
- Mainly traditional signal processing algorithms, but more data-driven training, e.g. PDP, NN intra, NSPT

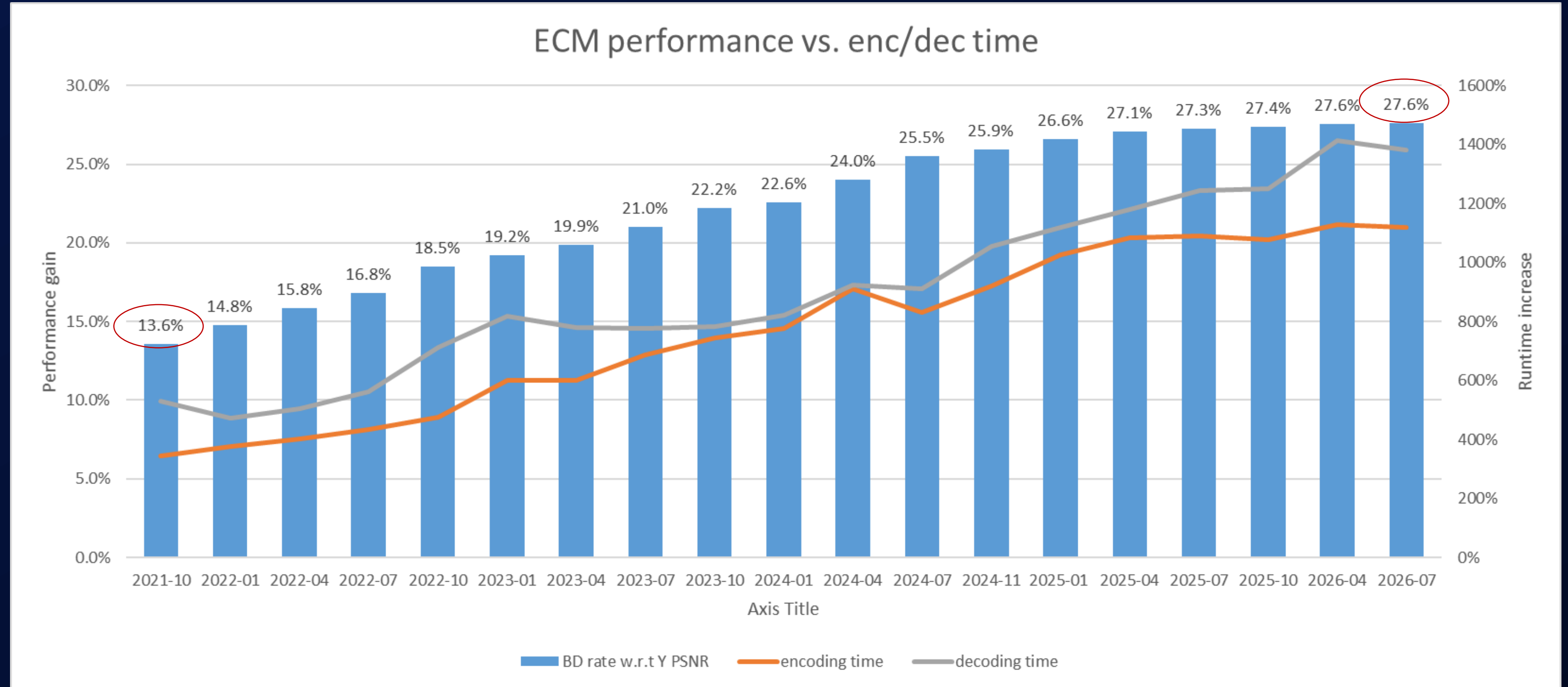
ECM evolution: performance vs. runtime

Random Access (RA)
config. of ECM
common test
conditions

YUV4:2:0 10b coding,
covering SD, HD, UHD
resolutions

BD rate savings (%) in
terms of Y PSNR

Encoding and
decoding time relative
to VVC ref. sw. VTM



Neural Network-based Video Coding (NNVC)

- Besides ECM (more focused on non-NN coding tools), JVET also has been exploring neural network-based video coding since 2020
- NNVC software uses fixed point implementation for inference and therefore results reproducible
- NNVC-17.1 default (NN intra + low complexity loop filters) achieves 8.2% gain over VTM with 20x decoder runtime on CPU
- Other variations (other loop filter flavors, deep reference frame, etc) achieve lower/higher gains with different complexity implications

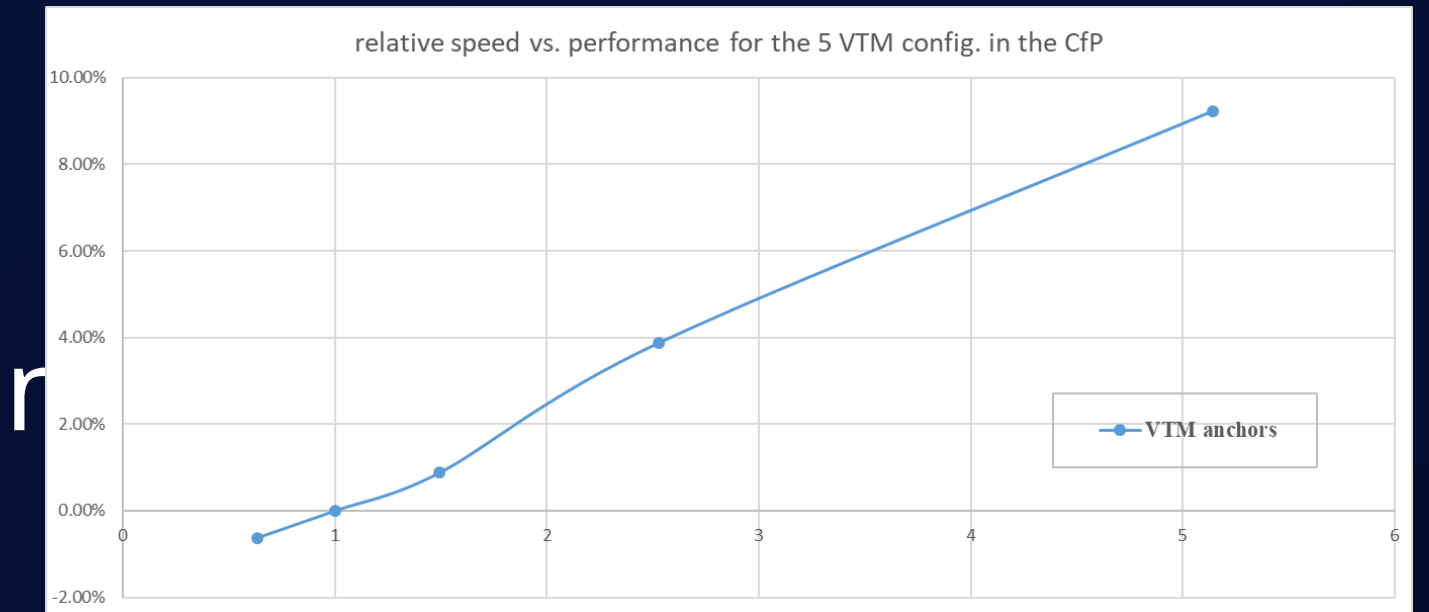
	Luma BD-rate (%)	CPU EncT (times)	CPU DecT (times)	kMAC/pixel	Params (M)
NN-Intra + LOP (default)	−8.2	1.2	20	21.3	1.5
NN-Intra + VLOP	−6.1	1.2	15	9.9	1.3
NN-Intra + HOP	−14.1	3.4	1180	471	2.7
Default + L-DRF	−9.6	1.3	230	83	5.2
Default + H-DRF	−10.6	1.7	869	477	7.7
E2E AI for intra picture	−1.7	1.0	5.3	23	36.4

Towards the next gen. codec standard

- Observing the significant progress in coding efficiency gain that JVET has made, 20 companies proposed timeline and requirements for the next gen codec standard in October 2024
- In January 2025, ITU-T SG21 and ISO/IEC JTC 1 SC 29 jointly held “Workshop on Future video coding – advanced signal processing, AI and standards” in Geneva, Switzerland
- In July 2025, the joint Call for Evidence was issued, 5 responses were received in October, with subjective testing showing sufficient evidence to move forward with the next gen. project
- ECM- and NNVC-based responses
- In July 2026, the joint Call for Proposals was formally issued

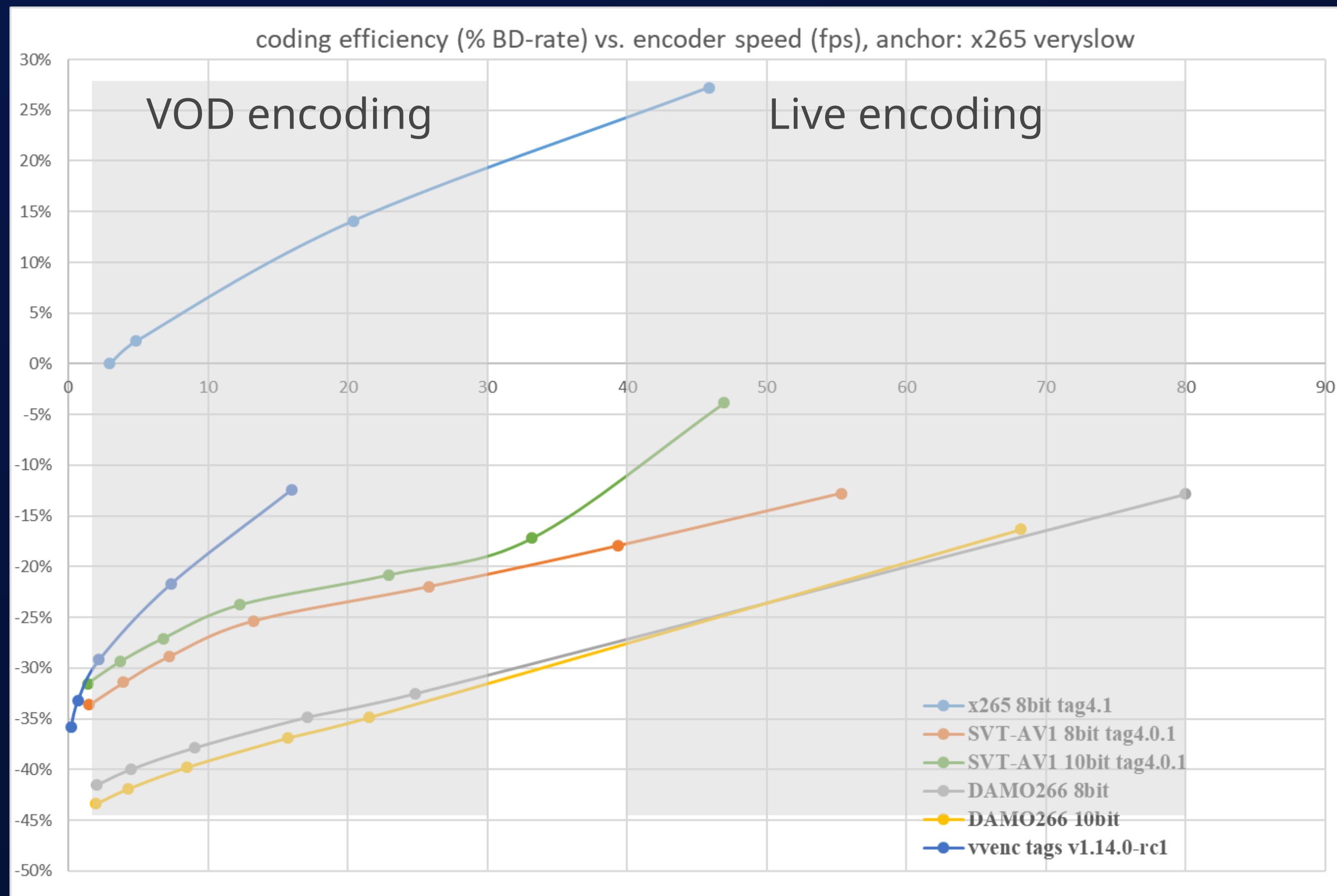
The Joint CfP

- Categories:
 - Improved compression
 - Improved compression with runtime-constrained encoding
 - Additional functionality
- First time to include runtime constrained encoding, 5 VTM configurations for different encoding runtime
 - 1x (default VTM), 1.5x, 2/3x, 0.4x, 0.2x
- In September, JVET announced 32 registered CfP responses from 33 companies
- Tentative timeline for the next gen. project
 - 2027-01 Initial test model selection process begins
 - 2027-10 Initial test model selection completed (at latest)
 - 2029-10 Final standard completed



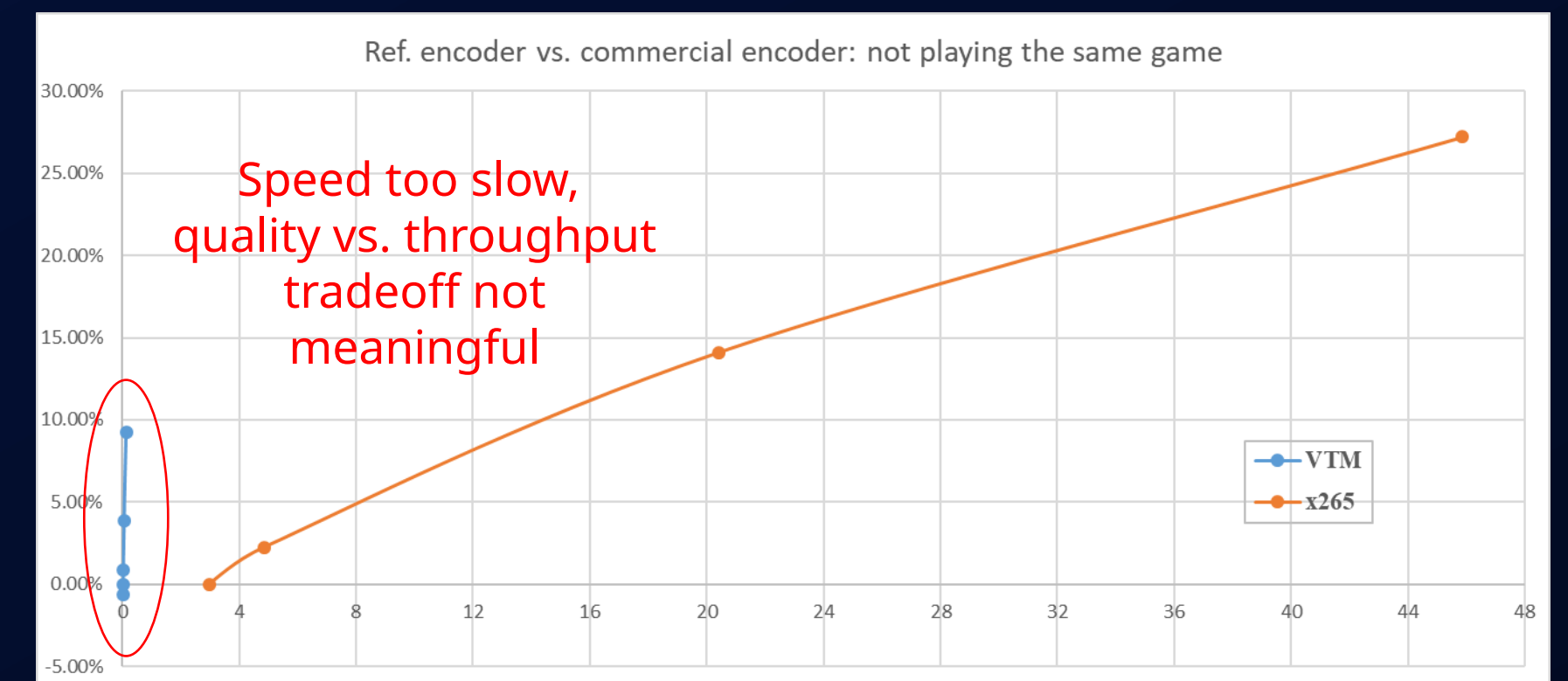
02 | Deploying an Advanced Codec

Encoders must serve a wide range of applications



- 31 HD sequences (movie, sports, UGC, games, JVET, etc.)
- Compute platform: AMD EPYC 7T83 64-Core Processor, ~2.6 GHz (2021)
- PSNR is the quality metric used to calculate BD-rate

- To serve different applications, commercial-grade encoders must cover a meaningfully **wide range of operating points** with **desirable quality vs. throughput tradeoffs**
- In contrast, ref. encoders only provide a “theoretical” performance ceiling



- VTM: BD-rate calculated against anchor VTM default config, performance and speed data from 8 HD sequences in CfP (JVET-AQ2021), mainstream platform
- x265: BD-rate anchor veryslow preset, 31 HD proprietary sequences, configs according to typical VOD and live transcoding applications, compute platform AMD EPYC 7T83 64-Core Processor, ~2.6 GHz (2021)
- PSNR is the quality metric used to calculate BD-rate

Decoders must be fast, lean and green

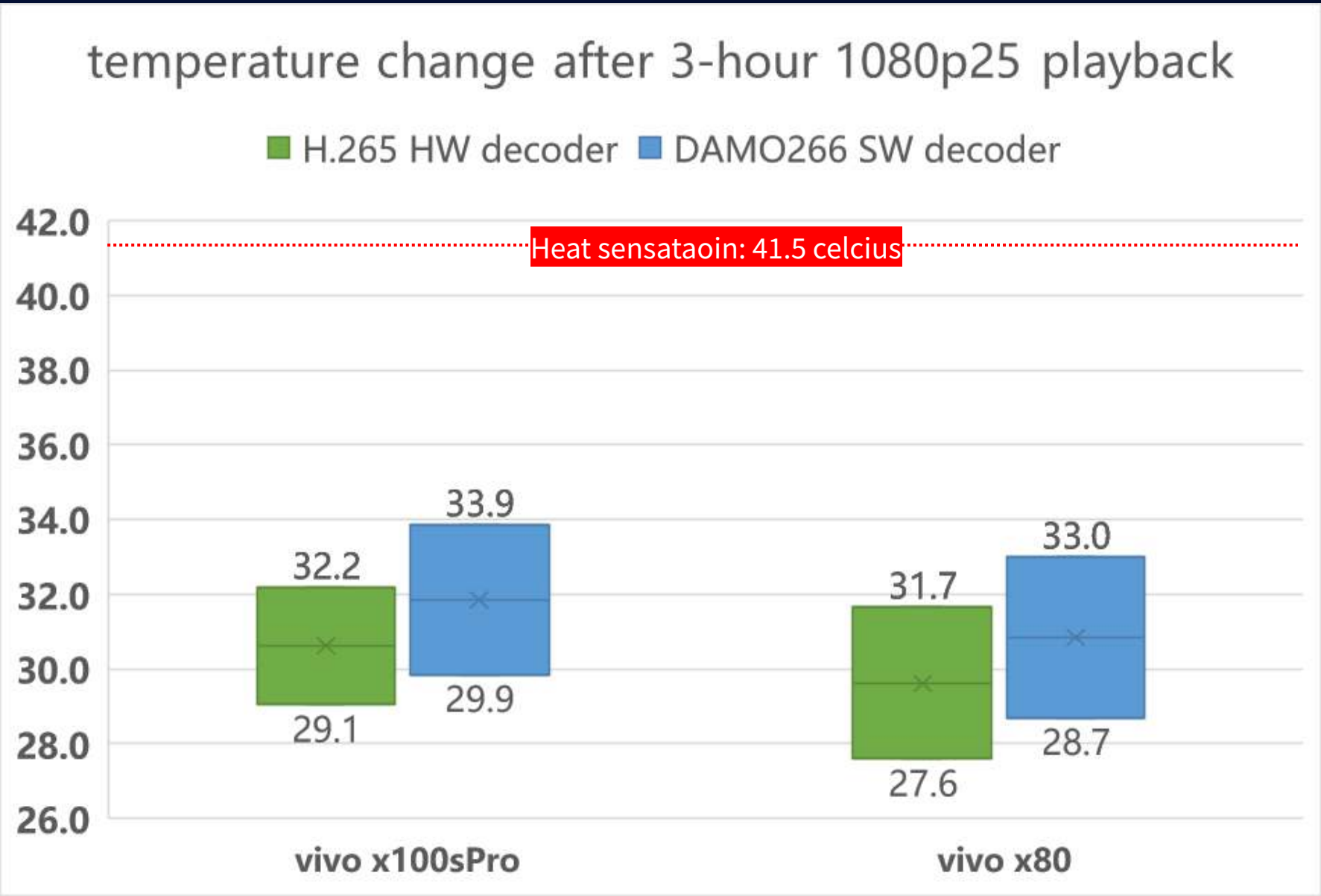
- For video playback, mobile is the most relevant target device for mass-scale consumer video applications
- As hardware decoding support on mobile has typically lagged standardization (gap of 3-5 years), **efficient software decoders** are the most essential enabler for meaningful deployment
- Software decoding: speed, memory and power consumption/battery drain
- Well-optimized software decoders are fast and lean, and can sustain handset battery for hours on end without causing overheating and user discomfort

		Speed (fps)	Memory (MB)
3 threads	DAMO266	334	122
	VVdeC	233	225

8 sequences, resolution 1920x1080, coded bit rate ~2 Mbps
Test device is Xiaomi 14 Pro, Qualcomm Snapdragon 8 Gen 3

Hours to drain 50% battery	1080p30 (3 threads)	1080p60 (3 threads)	4K30 (4 threads)
dav1d	5.8	3.6	3.6
DAMO266	5.3	3.3	3.0
VVdeC	4.0	2.3	2.5

VCAT-d: Open-source benchmark platform, measures frame decode time, CPU utilization and battery consumption under controlled conditions
 2 Netflix sequences coded with aligned quality (PSNR ≈ 39 dB)
 Test device is Xiaomi 14 Pro, Qualcomm Snapdragon 8 Gen 3, display at 50% brightness
 Charged to full, non-stop playback till 50% battery remains



vivo x100sPro, SOC MediaTek Dimensity 9300+
 vivo x80, SOC MediaTek Dimensity 9000"

03

AI and Video Coding

AI and video coding: putting the two together

- AI inside a video codec
 - Coding performance gain: a few percentages
 - Implementation hurdles: hardware area, CPU-GPU data transfer, memory footprint, battery drain, and so on
 - A reasonable tradeoff must be sorted out before NN tools can be deployed at mass scale, challenges for software-based decoding solutions in the near term



- AI in video workflows
 - As upstream and downstream tasks, AI is already deployed, and scale of deployment continues to ramp up quickly
 - AI content generation: very popular for content creators
 - AI content enhancement: super-resolution, temporal interpolation, etc.
 - AI content understanding: cross-platform content (e.g. summarization, landscape-to-portrait), end-user interaction, etc.
 - AI for 3D: 3D-fication of 2D content, etc.



Thank
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